

MORGAN S. THALBERG

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EXECUTIVE SUMMARY

Strategic and innovative Senior Solutions Architect with over 12 years of experience designing and implementing high-availability, scalable, and cost-effective enterprise systems. Expert at translating complex business requirements into robust technical architectures that drive digital transformation and operational efficiency. Proven track record of leading multi-cloud migrations (AWS, Azure, GCP), architecting event-driven microservices, and establishing enterprise-wide architectural standards. A collaborative leader who excels at stakeholder management, bridging the communication gap between C-suite executives and engineering teams. Deeply committed to security-by-design, infrastructure-as-code, and data-driven governance. Specialized in modernizing legacy monoliths, optimizing cloud spend, and implementing robust disaster recovery strategies for mission-critical applications.

STRATEGIC CORE COMPETENCIES

- **Enterprise Architecture:** TOGAF, Domain-Driven Design (DDD), Service-Oriented Architecture (SOA), Microservices, Serverless.
- **Cloud & Infrastructure:** Multi-Cloud Strategy (AWS/Azure), Kubernetes (EKS/AKS), Terraform, CloudFormation, Hybrid Cloud Connectivity.
- **Data & Integration:** Event-Driven Architecture (Kafka/RabbitMQ), GraphQL, RESTful APIs, ETL/ELT Pipelines, Snowflake, Databricks.
- **Security & Compliance:** Zero-Trust Architecture, OAuth2/OIDC, IAM Governance, SOC2/HIPAA Compliance, Data Encryption.
- **Business Strategy:** Cost Optimization (FinOps), Vendor Management, RFP/RFQ Evaluation, Roadmap Development, TCO Analysis.
- **Leadership:** Technical Mentorship, Cross-Functional Team Leadership, Agile/SAFe Methodologies, Stakeholder Management.

PROFESSIONAL EXPERIENCE

Principal Solutions Architect | Enterprise Cloud Tech | Denver, CO

June 2020 – Present

- **Strategic Cloud Transformation:** Led the end-to-end architecture for a \$50M digital transformation initiative, migrating 150+ legacy applications from on-premise data centers to a multi-cloud AWS/Azure environment.
- **Cost Optimization Leadership:** Established a FinOps practice that reduced annual cloud expenditure by \$3.2M (25% reduction) through right-sizing, reserved instances, and automated lifecycle management.
- **Enterprise Governance:** Authored the Global Architectural Standard for the organization, encompassing API versioning, security protocols, and CI/CD maturity models adopted by 800+ developers.

- **High-Scale Data Architecture:** Designed a real-time analytics platform using Snowflake and Apache Kafka, enabling sub-second business intelligence reporting for 5,000+ concurrent global users.
- **M&A Integration:** Orchestrated the technical integration of two acquired startups, successfully merging disparate identity providers (Okta/Active Directory) and consolidating cloud infrastructure within 6 months.
- **Disaster Recovery:** Architected a multi-region active-active disaster recovery solution that reduced RTO (Recovery Time Objective) from 4 hours to 15 minutes.

Senior Solutions Architect | Global Logistics Systems | Chicago, IL

January 2016 – May 2020

- **Supply Chain Modernization:** Designed a microservices-based tracking system using Node.js and Spring Boot, handling 10M+ daily events with 99.99% uptime.
- **Serverless Innovation:** Implemented an event-driven serverless architecture using AWS Lambda and Step Functions for automated order processing, reducing operational overhead by 40%.
- **API First Strategy:** Developed a public-facing Developer Portal and API Gateway (Apigee), enabling third-party integrations that increased partner-driven revenue by 15%.
- **Kubernetes Adoption:** Led the containerization strategy for the logistics core, migrating services to EKS and implementing GitOps with ArgoCD.
- **Stakeholder Management:** Collaborated with the VP of Operations to align technical roadmaps with global shipping expansion goals in the EMEA region.

Software Architect / Lead Developer | FinTech Solutions | New York, NY

June 2012 – December 2015

- **Monolith Decomposition:** Architected the transition of a legacy .NET monolith into a suite of scalable microservices, improving deployment frequency from monthly to daily.
- **Database Transformation:** Led the migration from a centralized SQL Server to a polyglot persistence model using PostgreSQL, Redis, and MongoDB.
- **Technical Leadership:** Managed a team of 12 developers, focusing on TDD, peer reviews, and automated integration testing.

ARCHITECTURE CASE STUDIES

Case Study: Global Legacy Modernization

The Challenge: The client faced high licensing costs and technical debt from a 15-year-old mainframe-based inventory system that couldn't handle holiday traffic spikes.

The Solution:

- **Pattern:** Utilized the "Strangler Fig" pattern to replace core modules incrementally.
- **Architecture:** Built a reactive event-mesh using Solace and Spring Cloud Stream to decouple the legacy core from new cloud-native microservices.
- **Result:** Successfully decommissioned the mainframe within 24 months. Scalability increased 10x, and maintenance costs dropped by 60%.

Case Study: Secure Multi-Tenant SaaS Platform

The Challenge: Developing a SaaS platform for healthcare providers that required strict data isolation, HIPAA compliance, and individual encryption keys for each tenant.

The Solution:

- **Design:** Implemented a "Cell-Based Architecture" where each tenant's data was isolated at the infrastructure level using AWS Nitro and KMS (Key Management Service).
- **Governance:** Automated compliance checks using AWS Config and OPA (Open Policy Agent) to ensure no non-compliant resources could be deployed.
- **Result:** Passed three independent security audits with zero findings. Onboarded 50+ enterprise healthcare clients in the first year.

TECHNICAL DEEP DIVES

Enterprise Integration Patterns (EIP)

Expertise in designing complex integration flows:

- **Messaging:** Asynchronous communication styles, dead-letter queues, and idempotent consumers.
- **API Management:** Rate limiting, circuit breakers (Resilience4j), and comprehensive observability (OpenTelemetry).

Security & Identity Architecture

- **Identity:** Deep knowledge of OIDC/SAML, SCIM for user provisioning, and Multi-Factor Authentication (MFA) implementation patterns.
- **Infrastructure Security:** VPC Peering, Transit Gateways, and WAF (Web Application Firewall) tuning.

DevOps & Automation (IaC)

- **Tools:** Terraform, Pulumi, Ansible.
- **Philosophy:** Immutable infrastructure and "Build it, Run it" cultural transformation.

EDUCATION & CERTIFICATIONS

Education

Master of Science in Information Systems Architecture

Carnegie Mellon University | 2012

Bachelor of Science in Computer Science

Georgia Institute of Technology | 2010

Professional Certifications

- AWS Certified Solutions Architect – Professional
- Microsoft Certified: Azure Solutions Architect Expert
- Google Cloud Professional Cloud Architect
- TOGAF 9 Certified
- Certified Kubernetes Administrator (CKA)

THOUGHT LEADERSHIP & GOVERNANCE

- **Architecture Review Board (ARB):** Chair of the internal ARB, reviewing 20+ major project designs quarterly.
- **Speaker:** "The Reality of Multi-Cloud Architecture" – Gartner IT Symposium, 2023.
- **Blogger:** "The Pragmatic Architect" – Monthly column focusing on TCO and technical debt.

DETAILED PROJECT PORTFOLIO

Project: "EdgeLink" - IoT Gateway

Designed the architecture for an edge-computing gateway that processed sensor data locally before syncing filtered results to the cloud, reducing bandwidth costs by 80%.

Project: Automated Compliance Engine

Built a Python-based tool that scans cloud environments against NIST and CIS benchmarks, automatically remediating security groups and unencrypted buckets.

CAREER TIMELINE

- **2010-2013:** Software Engineering - Focus on Java/Spring and SQL.
- **2014-2017:** System Design - Transition to Microservices and early Cloud.
- **2018-2021:** Solutions Architecture - Large-scale migrations and DevSecOps.
- **2022-Present:** Principal / Executive Level - Strategy, FinOps, and Governance.